Grades will be notified at

7:00pm of Tuesday, July 5, 2022

Question 1 (5 points)

In the closed economy Sigma, prices are constant and the central bank chooses, and keeps constant at the chosen level, the money supply M. The economy is described by an otherwise standard IS-LM model, with a consumption function that depends positively on disposable income, investment that depends positively on income (Y) and negatively on the interest rate (i), government purchases of goods and services and net taxes both exogenous ($G = \bar{G}$ and $T = \bar{T}$), and money demand increasing in Y and decreasing in i. Suppose that, starting from an initial equilibrium position, Sigma's government raises $\bar{G}$. How will private saving, public saving and national saving have changed, in the new equilibrium that Sigma will reach? And how would your answer change if you were told that, rather than the money supply, the country's central bank chooses the interest rate? Explain.

SKETCH OF ANSWER

Since the central bank chooses the money supply, the LM curve is positively sloped. The increase in $\overline{G}$ shifts the IS curve to the right. In the new equilibrium, income and the interest rate are higher. The direction in which investment, and therefore national saving, will change is therefore uncertain. However, government saving will certainly have decreased, and private saving increased (due to the increase in equilibrium income and therefore, for given net taxes, in disposable income).

If the central bank chooses the interest rate, the LM is horizontal. When the government raises $\overline{G}$, income goes up, but the interest rate does not change. As before, government saving decreases and private saving increases. In the new equilibrium, however, national savings will certainly be higher, as will investment (Y is higher, i unchanged).

Question 2 (5 points)

Using the diagram "supply of goods-demand for goods" employed to analyze the determination of the real interest rate consistent with goods markets equilibrium in the medium run, r_n, explain if you agree, or do not agree, with the following statement:
"*In an economy in which consumption depends positively on disposable income and negatively on the real interest rate, and in which investment and government spending are exogenous ($I = \bar{I}$ and $G = \bar{G}$), a decrease in investment is neutral in the medium run; in fact, it does not change the natural level of output, Y_n, as this latter depends on factors that have nothing to do with $\bar{I}$*".
In your answer, also explain which of the curves in the graph are affected by the change under consideration, and how.

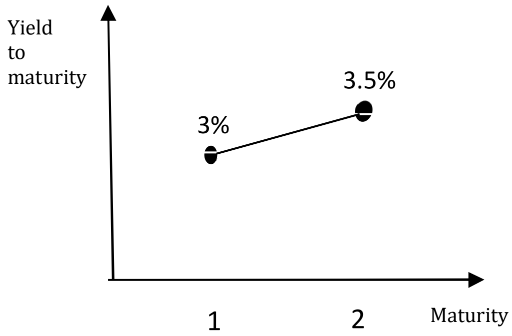
[NB: when answering, make sure to refer to the "supply of goods-demand for goods" graph; answers with no explicit reference to that graph and its implications will receive zero points].

SKETCH OF ANSWER

False. If investment falls from $\bar{I}$ to $\bar{I}' < \bar{I}$, aggregate demand will be lower for any level of the real interest rate, and the demand curve in the "supply of goods-demand for goods" graph will therefore shift to the left. In the move from the initial equilibrium to the new equilibrium, r_n falls by the amount needed to raise consumption exactly by the amount by which autonomous investment has gone down. It follows that, even though it does not lead to a change in the natural level of output, which depends on supply-side factors only, or in the overall level of the aggregate demand for goods, the change in $\bar{I}$ is not neutral in the medium run, as it leads to a change in the composition of aggregate demand.

Question 3 (5 points)

In country Zeta, only one-year and two-year bonds exist. Investors, who do care about risk, ask for a risk premium x to hold the two-year bond (which is risky, as they do not know the price at which they will be able to sell it in a year).
Suppose that, in the current period (time t), Zeta's yield curve is the one in the figure below. Write down the name of the variables measured along the axes and compute the numerical value that the risk premium x takes on in this economy, knowing that investors expect the future yield on one-year bonds to be the same as the current one ($i_{1t} = i^e_{1,t+1}$).
[Hint: use the (approximate) relation among the yield to maturity of a two-year bond (i_{2t})*, the current* (i_{1t}) *and future expected* ($i^e_{1,t+1}$) *yields on one-year bonds, and the risk premium* (x) *implied by the arbitrage condition.]*



SKETCH OF ANSWER

When investors are risk averse, the arbitrage condition implies that, for i_{2t}, i_{1t} *and* $i^e_{1,t+1}$ *'small enough' (close to zero), one can write*

$$i_{2t} = \frac{1}{2}\left(i_{1t} + i^e_{1,t+1} + x\right). \qquad (*)$$

In this economy, $i_{2t} = 3.5\%$ *and* $i_{1t} = i^e_{1,t+1} = 3\%$. *Plugging these values into* $(*)$,

$$3.5\% = \frac{1}{2}(3\% + 3\% + x)\,.$$

Solving,

$$x = 1\%\,.$$

Question 4 (5 points)

In the closed economy Alpha, there are only one-year and two-year bonds. The current and the expected future inflation rates are both zero, so that the real and the nominal interest rates are equal. In period $t+1$ ("the future"), consumption depends only on disposable income at $t+1$, and investment on the interest rate and output at $t+1$ only. Turning now to the current period (t, "the present"), consumption is increasing in both current and expected future disposable incomes, $C_t = C(Y_t - \bar{T}_t, Y^e_{t+1} - \bar{T}^e_{t+1})$, while investment depends positively on current and expected future income levels, and negatively on the current and expected future levels of short-term interest rates. Finally, in both periods G and T are exogenous and, rather than the interest rate, Alpha's central bank chooses the money supply.

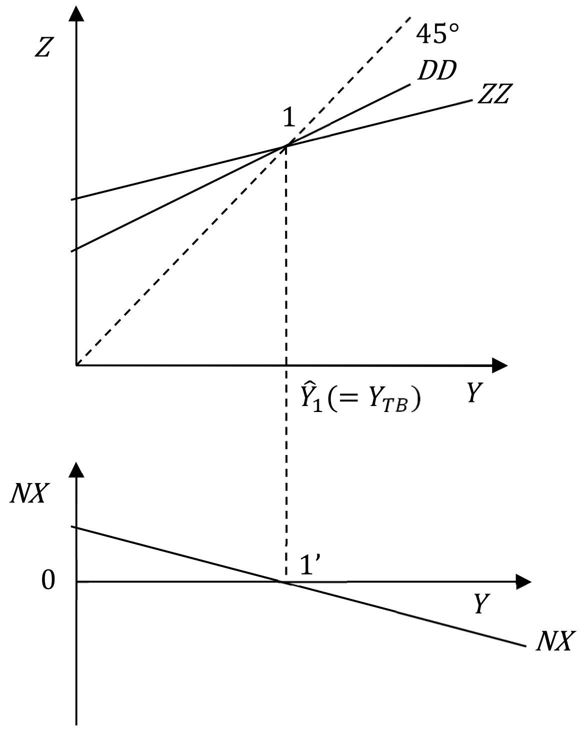
Assuming that, in the equilibrium initially prevailing at time *t*, the current and future expected yields on one-year bonds are $i_{1t} = 4\%$ and $i^e_{1,t+1} = 2\%$, write down the numerical values of the two points that Alfa's yield curve connects, and briefly comment on the resulting slope of that curve. Finally, suppose that market participants start expecting that, at time $t+1$, monetary policy will become more restrictive than initially assumed. Discuss if and how these expectations on the future monetary policy stance will affect i_{2t}, the time *t* yield on two-year bonds prevailing in Alpha, and the slope of the time *t* Alpha's yield curve. Will that slope increase, decrease, or remain constant? Explain.

SKETCH OF ANSWER

In this economy where only one-year and two-year bonds exist, the two points that the time t yield curve connects are $i_{1t} = 4\%$ *and the yield to maturity of two-year bonds issued at t,* $i_{2t} = \frac{1}{2}(i_{1t} + i^e_{1,t+1}) = \frac{1}{2}(4\% + 2\%) = 3\%$. *Coherently with the expectations of a decrease in short term interest rate that characterize this economy, the yield curve has therefore a negative slope. Since the central bank chooses the money supply, the LM curve will be positively sloped in both periods. A restrictive monetary policy implemented at time t+1 will cause a parallel shift to the left of the LM, leading to a higher short term interest rate and a lower equilibrium income in that time period. Both changes will shift the IS curve for time t to the left, thus lowering not just* Y_t*, but also* i_{1t}*, the first of the two points connected by the yield curve. As for* i_{2t}*, the time t yield on two-year bonds, it is not possible to say anything definite about the direction in which it will vary: the current short term rate falls, but the future one increases. Finally, depending on the extent by which* i_{1t}*,* $i^e_{1,t+1}$*, and therefore* i_{2t}*, will change, the yield curve could become positively sloped, flat, or keep its negative slope. Even if it remains negatively sloped, however, its slope will increase – it will become flatter that the initial curve. The reason is that, even if* i_{2t} *falls,* $\Delta i_{2t} = \frac{1}{2}(\Delta i_{1t} + \Delta i^e_{1,t+1})$*,* $\Delta i_{1t} < 0$ *and* $\Delta i^e_{1,t+1} > 0$ *imply that it will fall less than* i_{1t}*. This increase in the slope of the yield curve is in line with the anticipation of a future short rate higher than originally expected.*

Question 5 (5 punti)

Consider an economy open to trade with the rest of the world, where prices are constant and in which only the goods market exists. Net exports depend positively on the real exchange rate, so that the Marshall-Lerner condition does not hold. Assuming that, starting from the initial equilibrium (1,1') in the figure, domestic prices P permanently decrease ($\Delta P < 0$), discuss how the slope and/or the position in the plane of each of the three curves ZZ, DD and NX will vary. In the new equilibrium that the economy will reach, how will income and net exports have changed? Which kind of fiscal policy (expansionary/restrictive) and/or exchange rate policy (increase in E / decrease in E) could bring back production to the initial level ($\hat{Y}_1$) and net exports in balance ($NX = 0$)? Explain.



SKETCH OF ANSWER

A decrease in domestic prices lowers the real exchange rate $\varepsilon \equiv (EP)/P^*$ (a real depreciation). Since the Marshall-Lerner condition does not hold, the fall in ε decreases net exports for any given Y, so that the ZZ and the NX curves shift downwards. Even though equilibrium income is lower (in the first graph, the ZZ now crosses the 45° line for a lower Y), in the new equilibrium net exports, initially balanced, will now be negative. In fact, the new equilibrium income is larger than the new Y_{TB} – the level of production for which the ZZ and the DD curves intersect, and therefore for which $NX = 0$. An expansionary fiscal policy could of course bring back income to the (higher) initial equilibrium level; however, precisely for this reason, it would also make the trade deficit worse. In order to restore the initial level of income, and a zero trade balance, it is necessary to intervene on the exchange rate. More specifically, the government should manage to engineer an appreciation of the nominal exchange rate E, thus offsetting the impact of the decrease in domestic prices on ε, and bringing back the ZZ and the NX curves to their initial position.

Question 6 (5 points)

Consider Delta, an economy with constant prices under a fixed exchange rate regime described by a standard open economy "IS-LM-interest parity" model. Delta's central bank is determined to keep the exchange rate constant at the chosen level $E = E_1$, and individuals expect it to do so also in the future, so that $E = E_1 = E^e$. Suppose that, starting from an initial equilibrium position, the foreign interest rate goes up ($\Delta i^* > 0$) and that, at the same time, foreign income goes down ($\Delta Y^* < 0$). Which of the three curves – IS, LM, interest parity – will be affected by these two contemporaneous changes, and why? In the move from the initial equilibrium to the new one that the economy will reach, how will Delta's equilibrium levels of production, interest rate, exchange rate, consumption, investment and net exports will change? And what about money supply and money demand? Explain.

SKETCH OF ANSWER

- *The increase in the foreign interest rate leads to an upward shift in the line representing the interest parity condition − the new line will now go through, as it has to do by construction, the point ($E = E^e, i = i^{*\prime}$), where $i^{*\prime}$ is the new level of the foreign interest rate.*
- *By decreasing net exports, the fall in foreign income will shift the IS curve to the left.*
- *Absent any intervention by the domestic central bank, the exchange rate would depreciate (it would fall below E_1). To keep it at E_1, the central bank will have to intervene, increasing the domestic interest rate by the same amount by which the foreign one has gone up. This intervention will shift the LM curve upwards, till it becomes horizontal at the new, higher level of the domestic interest rate $i = i^{*\prime}$.*
- *In the new equilibrium, income is lower (due to the leftward shift of the IS caused by the decrease in foreign income and to the increase in the domestic rate i that the central bank has implemented in order to keep the exchange rate constant), i higher, E unchanged, C lower (Y is lower), I has gone down (Y is lower, i higher). The direction in which net exports have changed is uncertain (E has not changed, but domestic and foreign income levels have gone down, pushing NX in opposite directions). Finally, in the new equilibrium money supply is lower (to keep E unchanged, the central bank has raised the domestic rate); the same is true about money demand ((Y is lower, i higher).*